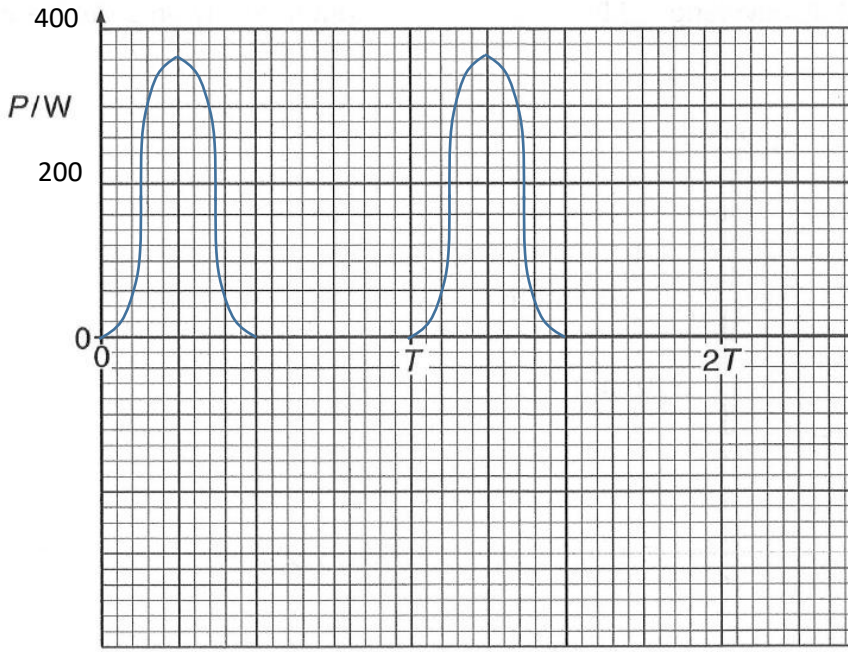
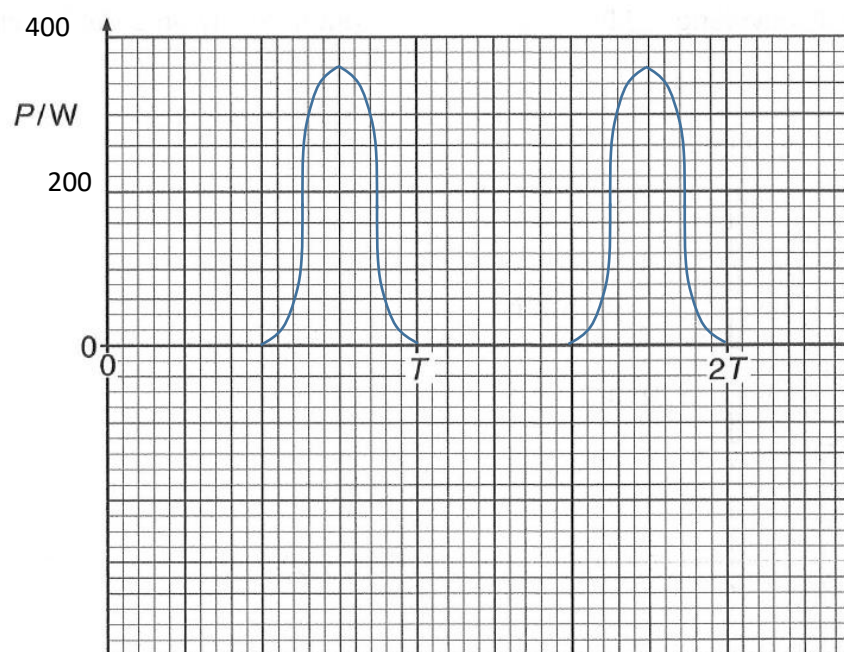


			ks
7	(a)	(i) $\frac{N_S}{N_P} = \frac{V_S}{V_P}$ $\frac{450}{2700} = \frac{V_s}{240}$ $V_s = 40V$ Peak voltage = $40 \times \sqrt{2}$ $= 56.5$ $= 57 V$	M1 A1
		(ii) r.m.s voltage across the load = $57 / 2$ $= 28 V$ (or $29 V$)	M0 A1
		(iii) $P = \frac{V^2}{R} \rightarrow P \propto \frac{1}{R}$ $\frac{P_Y}{P_X} = \frac{R_Y}{R_X} \rightarrow \frac{P_Y}{90} = \frac{1}{2}$ $P_Y = 45W$ $(0.90)P_{primary} = (P_{secondary})$ $(0.90)/V = 90 + 45$ $(0.90)(240)/I = 90 + 45$ $I = 0.63 A$	C1 C1 A1
	(b)	Half-wave rectified (see below for diagram) Peak power 360 W (unit to be written) Either: 	B1 B1

OR:



Question			Solution	Marks
8	(a)		• photon gives energy to electron (in an inner shell) or electron (in an	B1